LAB 4 Individual Deliverable 1

Single Segment Trajectory Optimization

Item	Value
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Problem

Minimize the velocity cost J = integral from 0 to 1 of $(P'(t))^2$ dt, subject to P(0) = 0 and P(1) = 1. Since P(0) = 0, the constant coefficient is zero in every polynomial below.

1. Degree-one polynomial

For $P(t) = p_1 t$, the terminal constraint gives $p_1 = 1$.

```
P*(t) = t
J* = integral_0^1 1^2 dt = 1
```

Thus the degree-one optimum is P*(t) = t and its minimum cost is 1.

2. Degree-two polynomial

Let $p = [p_1, p_2]^T$ and $P(t) = p_2 t^2 + p_1 t$. Then $P'(t) = p_1 + 2p_2 t$.

```
J = integral_0^1 (p1 + 2 p2 t)^2 dt
  = p1^2 + 2 p1 p2 + (4/3) p2^2
  = p^T Q p,   Q = [[1, 1], [1, 4/3]]

P(1) = p1 + p2 = 1,   so A = [1  1],  b = 1.
```

Using the equality constraint $p_1 = 1 - p_2$, the reduced cost is $J = 1 + p_2^2/3$. It is minimized by $p_2 = 0$ and $p_1 = 1$.

```
p* = [1, 0]^T,   P*(t) = t,   J* = 1.
```

Allowing degree two does not lower the cost because the minimum-velocity curve joining these two positions is already a line.

3. Degree-three polynomial

For $p = [p_1, p_2, p_3]^T$ and $P(t) = p_3 t^3 + p_2 t^2 + p_1 t$:

```
Q = [[1,   1,   1],
     [1, 4/3, 3/2],
     [1, 3/2, 9/5]],
A = [1  1  1],   b = 1.
```

The KKT equations, or direct minimization subject to $p_1 + p_2 + p_3 = 1$, give $p^* = [1, 0, 0]^T$. Therefore P*(t) = t and J* = 1.

This agrees with Euler-Lagrange: for r = 1, $d^2P/dt^2 = 0$. The solution satisfying the two position constraints is the linear curve P(t) = t.

4. Added final-velocity constraint

Now impose P'(1) = -2 in addition to the two position constraints.

4(a). Degree-two solution

```
P(t) = p2 t^2 + p1 t
p1 + p2 = 1
p1 + 2 p2 = -2
Therefore p1 = 4, p2 = -3.
```

Hence $P^*(t) = 4t - 3t^2$. Its cost is:

```
J* = integral_0^1 (4 - 6t)^2 dt = 4.
```

4(b). Degree-three solution

The two equality constraints are $p_1 + p_2 + p_3 = 1$ and $p_1 + 2p_2 + 3p_3 = -2$. Let $s = p_3$. Then $p_2 = -3 - 2s$ and $p_1 = 4 + s$.

```
J(s) = 4 + s + (2/15)s^2.
dJ/ds = 1 + (4/15)s = 0  =>  s = -15/4.

p* = [1/4, 9/2, -15/4]^T
P*(t) = (1/4)t + (9/2)t^2 - (15/4)t^3
J* = 17/8.
```

The degree-three model has lower cost (17/8 instead of 4) because it has one extra degree of freedom while satisfying the same three endpoint constraints.

Reference

MIT VNAV Lab 4 Exercises, Individual Deliverable 1. https://vnav.mit.edu/labs_2023/lab4/exercises.html